Experiment 7: Breadth First Search (BFS)

Aim

To write a Python program to implement the Breadth First Search (BFS) algorithm for graph traversal.

Procedure

1. Create a graph using an adjacency list.
2. Choose the starting node.
3. Create a queue and insert the starting node.
4. Mark the starting node as visited.
5. Remove one node from the queue.
6. Visit all its unvisited neighboring nodes and insert them into the queue.
7. Repeat until the queue becomes empty.
8. Display the BFS traversal.

Source Code

main.py Visualize Run ▶ Output

```python
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()
queue = deque()

start = input("Enter starting node: ")

visited.add(start)
queue.append(start)

print("BFS Traversal:")

while queue:
    node = queue.popleft()
    print(node, end=" ")

    for neighbour in graph[node]:
        if neighbour not in visited:
            visited.add(neighbour)
            queue.append(neighbour)
```

```
Enter starting node: A
BFS Traversal:
A B C D E F
```

Result

Thus, the Breadth First Search (BFS) algorithm was implemented successfully.